

Certified Assessment 3

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Reference code: MY-649

1. AI coding tool used

[Claude](#)

2. Exact AI prompts submitted

Prompt 1 : please write a coding leaflet application to locate the coordinate of places in ASEAN . Please use the attached document for marking a plotting. Label each location.

The ASEAN Shp data was attached to this prompt.

Prompt 2 : Hey, why is the searching not working? I cant click any countries either. also each place should have columns name, latitude, longitude and country

3. Generated Source Code

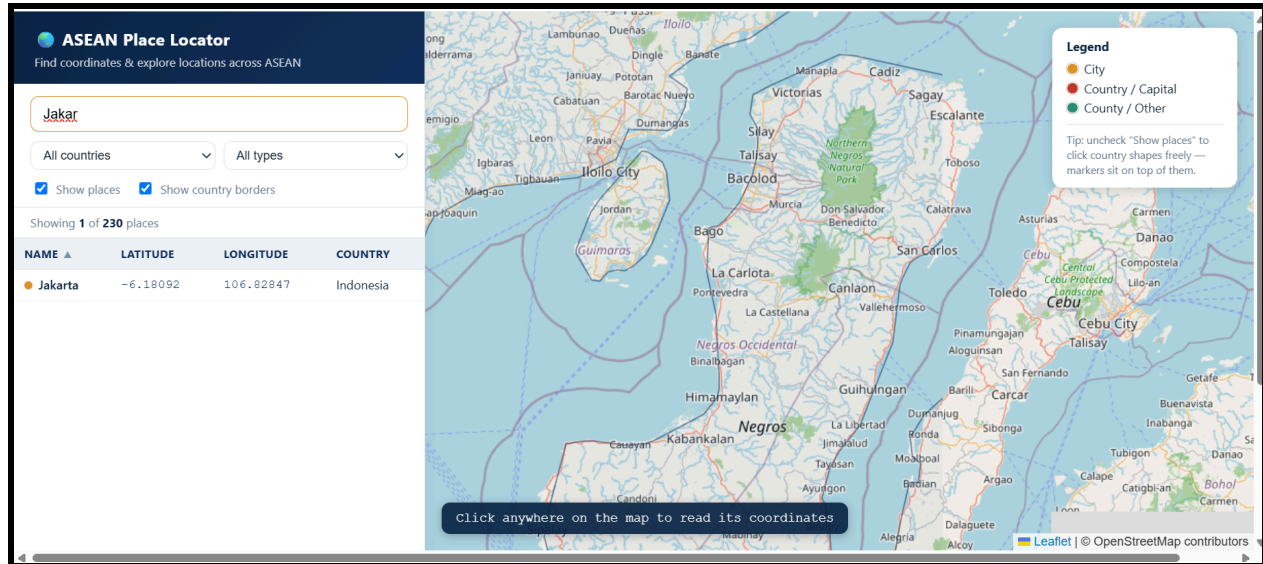
Link to source code :

<https://colab.research.google.com/drive/1BF3U4USnYcjjSTOQm0w1B5BxPbp8WzP7?usp=sharing>

4. Corrections, Modifications or Improvements made to the code

- The initial code generated did not allow the map to be searchable or clickable. Therefore, it wasn't really functionable. The countries were all also in their own native language. I reprompted Claude to fix that.

5. Screenshots of completed interactive map



6. Brief Explanation of Development, Testing and Troubleshooting process

a. Data inspection: the shapefiles were read using the pyshp library to confirm geometry type, attribute fields, feature counts, and coordinate reference system (read from each .prj file).

b. Format conversion: Leaflet requires geographic (WGS84 / EPSG:4326) GeoJSON. The country boundary layer was reprojected from Web Mercator to WGS84 using pyproj, then both layers were exported to GeoJSON (data/asean_countries.geojson and data/asean_places.geojson).

c. Application build: an index.html / style.css / script.js structure was created using [Leaflet.js](https://leafletjs.com/) (loaded via CDN), an OpenStreetMap base tile layer, a sidebar with layer toggles and a legend, and an information panel that updates on hover/click.

d. Feature implementation: country polygons are styled with a categorical colour per country and a hover-highlight effect; place points are rendered as circle markers coloured by place type (city / county / country) and sized by population where reported; both layers carry bound pop-ups and tooltips showing attribute information; a layer control lets the user toggle each layer on or off; the map automatically fits its initial view to the combined extent of the loaded data once both layers have loaded.

e. Testing: the application was served locally and tested in a headless Chromium browser (via Playwright) to confirm that both GeoJSON layers loaded, rendered with correct styling, and that pop-ups displayed the correct attribute values on click. Screenshots were captured to verify visual output.